POST-STA STATIC TIMING ANALYSIS

VSDBabySoC • Sky130HD • 16 PVT corners

Design	VSDBabySoC	**Technology / PDK**	Sky130HD / sky130_fd_sc_hd
Analysis tool	OpenROAD / OpenSTA	**Database**	6_final.odb
Constraints	6_final.sdc	**Parasitics**	Global-routing estimates
Corner coverage	16 PVT corners (FF / SS / TT)	**Clock handling**	Propagated clocks

Report basis. This document preserves the original 34-page STA report in full. The added overview and plots are derived from the 16-corner summary values in that report; no timing values have been changed.

Important scope note. The source report states that parasitics were estimated at global-routing level. This report therefore describes STA using global-routing parasitic estimates, not extracted-SPEF-based post-route STA.

Prepared from the supplied VSDBabySoC STA report

Executive Summary — All 16 PVT Corners

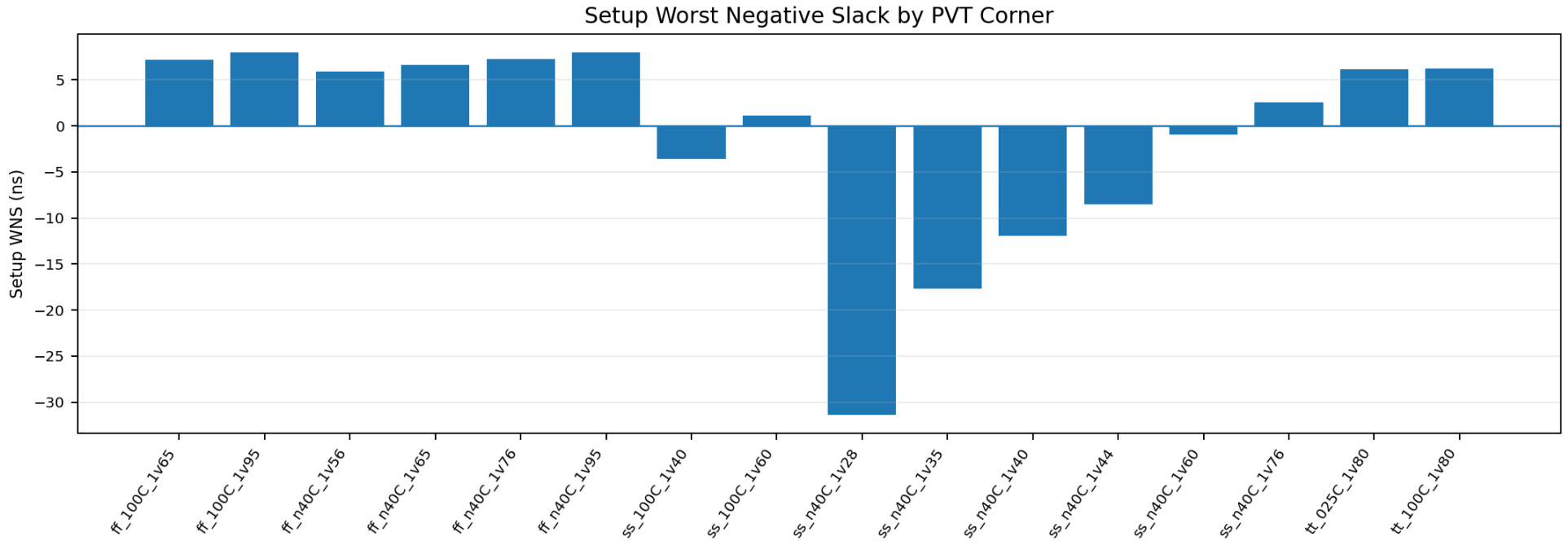
The table below reproduces the source report's corner-level setup and hold metrics. Negative setup WNS/TNS values are retained and indicate setup timing violations at those corners.

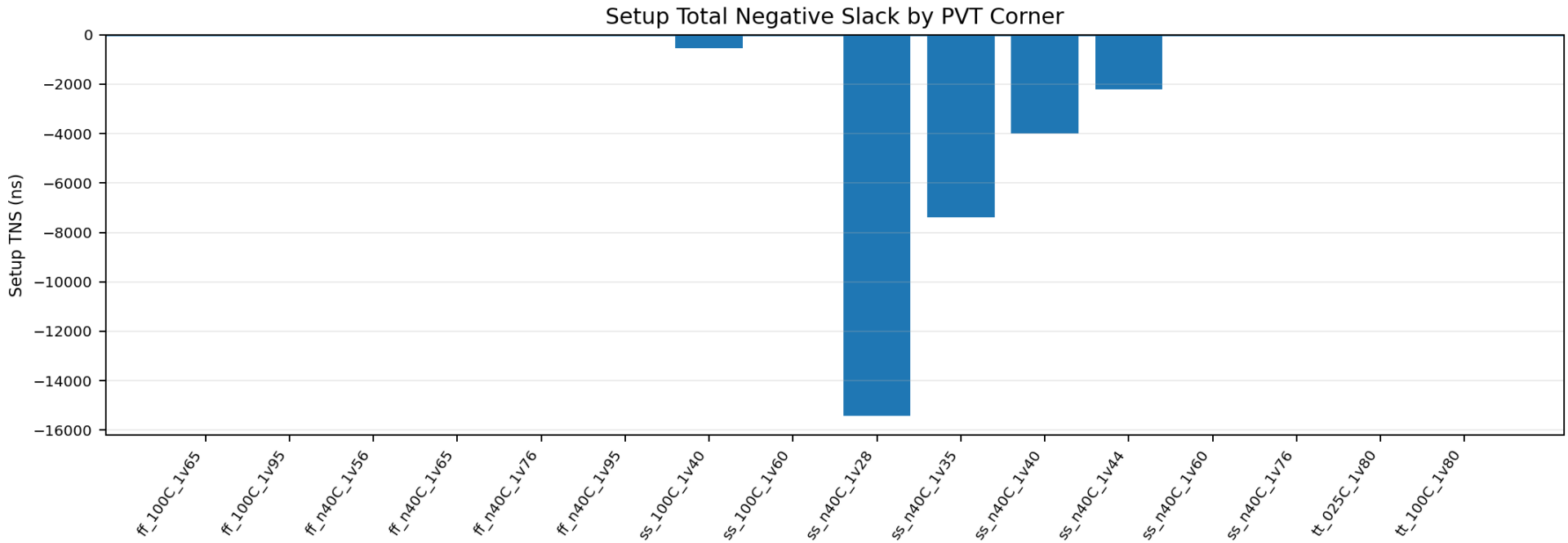
#	Corner	Process	Temperature / Voltage	Setup WNS (ns)	Setup TNS (ns)	Hold worst slack (ns)	Hold TNS (ns)
1	ff_100C_1v65	FF	100°C, 1.65 V	7.14	0.00	0.27	0.00
2	ff_100C_1v95	FF	100°C, 1.95 V	8.00	0.00	0.21	0.00
3	ff_n40C_1v56	FF	–40°C, 1.56 V	5.89	0.00	0.32	0.00
4	ff_n40C_1v65	FF	–40°C, 1.65 V	6.63	0.00	0.28	0.00
5	ff_n40C_1v76	FF	–40°C, 1.76 V	7.28	0.00	0.25	0.00
6	ff_n40C_1v95	FF	–40°C, 1.95 V	8.01	0.00	0.21	0.00
7	ss_100C_1v40	SS	100°C, 1.40 V	-3.57	-543.03	0.95	0.00
8	ss_100C_1v60	SS	100°C, 1.60 V	1.14	0.00	0.67	0.00
9	ss_n40C_1v28	SS	–40°C, 1.28 V	-31.43	-15435.67	1.85	0.00
10	ss_n40C_1v35	SS	–40°C, 1.35 V	-17.65	-7396.50	1.38	0.00
11	ss_n40C_1v40	SS	–40°C, 1.40 V	-11.95	-3993.60	1.16	0.00
12	ss_n40C_1v44	SS	–40°C, 1.44 V	-8.49	-2218.38	1.02	0.00
13	ss_n40C_1v60	SS	–40°C, 1.60 V	-0.98	-16.12	0.69	0.00
14	ss_n40C_1v76	SS	–40°C, 1.76 V	2.58	0.00	0.52	0.00
15	tt_025C_1v80	TT	25°C, 1.80 V	6.15	0.00	0.33	0.00
16	tt_100C_1v80	TT	100°C, 1.80 V	6.24	0.00	0.34	0.00

Summary: 6 of 16 corners have negative setup WNS and negative setup TNS. All 16 reported hold worst-slack values are positive, and hold TNS is 0.00 ns in every corner. This summary does not label the design as meeting setup timing across all corners.

Timing Metric Plots

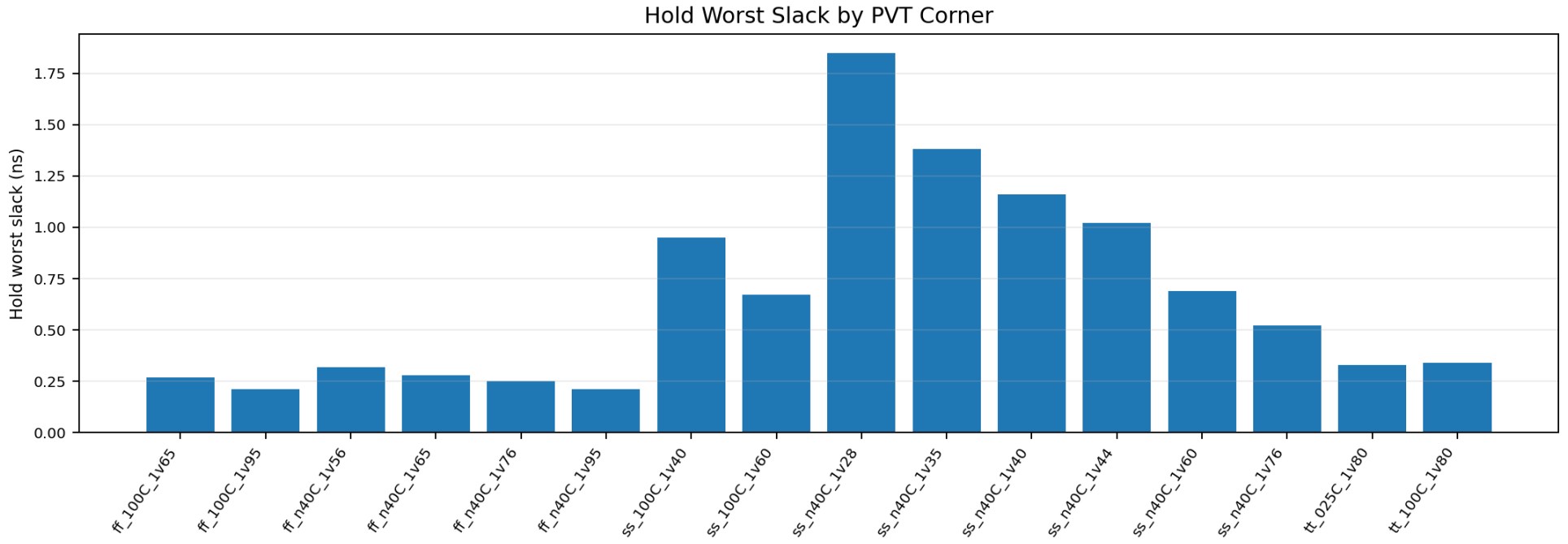
These plots visualize the reported summary values by corner. They are summary plots generated from the source table, not waveform or timing-path diagrams from the OpenROAD tool.




Setup Total Negative Slack by PVT Corner
Setup TNS (ns)
0
−2000
−4000
−6000
−8000
−10000
−12000
−14000
−16000
ff_100C_1v65
ff_100C_1v95
ff_n40C_1v56
ff_n40C_1v65
ff_n40C_1v76
ff_n40C_1v95
ss_100C_1v40
ss_100C_1v60
ss_n40C_1v28
ss_n40C_1v35
ss_n40C_1v40
ss_n40C_1v44
ss_n40C_1v60
ss_n40C_1v76
tt_025C_1v80
tt_100C_1v80

Hold Timing Overview

Hold worst slack is positive at all 16 corners in the supplied summary. Hold TNS is reported as 0.00 ns for every corner.



Methodology stated in source: OpenROAD v2.0-25794-g7cbd77e248; design database 6_final.odb; constraints 6_final.sdc; propagated clocks; parasitics estimated with `estimate_parasitics -global_routing`.

The original report follows unchanged as the detailed report section, including its original per-corner timing-path outputs and page formatting.

VSDBabySoC OpenROAD Static Timing Analysis

Sky130HD | All 16 PVT corners | Setup and hold summary

Methodology: OpenROAD v2.0-25794-g7cbd77e248; final routed database and SDC; corner-specific Sky130HD timing libraries; propagated clocks; parasitics estimated with `estimate_parasitics -global_routing`.

Corner	Process	Temperature / Voltage	Setup WNS (ns)	Setup TNS (ns)	Hold worst slack (ns)	Hold TNS (ns)
ff_100C_1v65	FF	100°C, 1.65 V	7.14	0.00	0.27	0.00
ff_100C_1v95	FF	100°C, 1.95 V	8.00	0.00	0.21	0.00
ff_n40C_1v56	FF	–40°C, 1.56 V	5.89	0.00	0.32	0.00
ff_n40C_1v65	FF	–40°C, 1.65 V	6.63	0.00	0.28	0.00
ff_n40C_1v76	FF	–40°C, 1.76 V	7.28	0.00	0.25	0.00
ff_n40C_1v95	FF	–40°C, 1.95 V	8.01	0.00	0.21	0.00
ss_100C_1v40	SS	100°C, 1.40 V	–3.57	–543.03	0.95	0.00
ss_100C_1v60	SS	100°C, 1.60 V	1.14	0.00	0.67	0.00
ss_n40C_1v28	SS	–40°C, 1.28 V	–31.43	–15435.67	1.85	0.00
ss_n40C_1v35	SS	–40°C, 1.35 V	–17.65	–7396.50	1.38	0.00
ss_n40C_1v40	SS	–40°C, 1.40 V	–11.95	–3993.60	1.16	0.00
ss_n40C_1v44	SS	–40°C, 1.44 V	–8.49	–2218.38	1.02	0.00
ss_n40C_1v60	SS	–40°C, 1.60 V	–0.98	–16.12	0.69	0.00
ss_n40C_1v76	SS	–40°C, 1.76 V	2.58	0.00	0.52	0.00
tt_025C_1v80	TT	25°C, 1.80 V	6.15	0.00	0.33	0.00
tt_100C_1v80	TT	100°C, 1.80 V	6.24	0.00	0.34	0.00

Interpretation: Six corners have negative setup WNS and negative setup TNS, indicating setup timing violations at those conditions. All 16 reported hold worst-slack values are positive, and hold TNS is 0.00 ns in every corner. Negative setup results are retained, not hidden.

Scope note: Parasitics were estimated at global-routing level. This is not extracted-SPEF-based post-route STA. Detailed per-corner setup/hold path reports follow this summary.

VSDBabySoC Static Timing Analysis

OpenROAD-generated reports for 16 Sky130HD timing library conditions

Design database: 6_final.odb
Constraints: 6_final.sdc
Parasitics: global-routing estimates
Each following section contains output captured from an individual OpenROAD run, including setup and hold summaries and timing paths.

Corner 1 of 16: ff_100C_1v65

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ff_100C_1v65.lib

===== SETUP SUMMARY =====
worst slack max 7.14
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2254    0.2254 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2629    0.4883 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1718    0.6602 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.6618 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2620    0.9239 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2493    1.1731 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.2192    1.3924 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.4982    1.8906 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.5653    2.4559 v _07716_/X (sky130_fd_sc_hd__mux4_2)
   0.1702    2.6261 ^ _07717_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1232    2.7494 v _07719_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0832    2.8326 ^ _08429_/Y (sky130_fd_sc_hd__nand2_1)
   0.1994    3.0320 v _08436_/Y (sky130_fd_sc_hd__a31oi_1)
   0.2654    3.2974 ^ _08462_/Y (sky130_fd_sc_hd__o211ai_1)
   0.3096    3.6070 v _08477_/Y (sky130_fd_sc_hd__a41oi_1)
   0.6907    4.2977 ^ _08478_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1331    4.4308 v _09004_/Y (sky130_fd_sc_hd__a21oi_1)
   0.0002    4.4311 v core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             4.4311   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.2254   11.2254 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2809   11.5063 ^ clkbuf_3_4__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1814   11.6877 ^ clkbuf_leaf_77_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0012   11.6889 ^ core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.6889   clock reconvergence pessimism
  -0.1223   11.5666   library setup time
             11.5666   data required time
---------------------------------------------------------
             11.5666   data required time
             -4.4311   data arrival time
---------------------------------------------------------
              7.1355   slack (MET)
===== HOLD SUMMARY =====
worst slack min 0.27
tns min 0.00
```

```
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2254    0.2254 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2629    0.4883 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1695    0.6578 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.6595 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2474    0.9069 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    0.9070 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             0.9070   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2254    0.2254 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2629    0.4883 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1695    0.6578 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    0.6593 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.6593   clock reconvergence pessimism
  -0.0227    0.6366   library hold time
             0.6366   data required time
---------------------------------------------------------
             0.6366   data required time
            -0.9070   data arrival time
---------------------------------------------------------
             0.2703   slack (MET)
```

Corner 2 of 16: ff_100C_1v95

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ff_100C_1v95.lib

===== SETUP SUMMARY =====
worst slack max 8.00
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.1896    0.1896 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2135    0.4031 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1385    0.5416 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.5433 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2082    0.7515 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2045    0.9560 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.1657    1.1217 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.4034    1.5250 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.4148    1.9399 v _07716_/X (sky130_fd_sc_hd__mux4_2)
   0.1345    2.0744 ^ _07717_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0965    2.1710 v _07719_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0655    2.2364 ^ _08429_/Y (sky130_fd_sc_hd__nand2_1)
   0.1633    2.3997 v _08436_/Y (sky130_fd_sc_hd__a31oi_1)
   0.2000    2.5997 ^ _08462_/Y (sky130_fd_sc_hd__o211ai_1)
   0.2446    2.8443 v _08477_/Y (sky130_fd_sc_hd__a41oi_1)
   0.5405    3.3848 ^ _08478_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0840    3.4688 v _09004_/Y (sky130_fd_sc_hd__a21oi_1)
   0.0002    3.4691 v core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             3.4691   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.1896   11.1896 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2281   11.4176 ^ clkbuf_3_4__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1448   11.5625 ^ clkbuf_leaf_77_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0012   11.5637 ^ core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.5637   clock reconvergence pessimism
  -0.0920   11.4717   library setup time
             11.4717   data required time
---------------------------------------------------------
             11.4717   data required time
             -3.4691   data arrival time
---------------------------------------------------------
              8.0027   slack (MET)
===== HOLD SUMMARY =====
worst slack min 0.21
tns min 0.00
```

```
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.1896    0.1896 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2135    0.4031 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1367    0.5398 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.5415 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.1962    0.7377 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    0.7378 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             0.7378   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.1896    0.1896 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2135    0.4031 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1367    0.5398 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    0.5413 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.5413   clock reconvergence pessimism
  -0.0173    0.5240   library hold time
             0.5240   data required time
---------------------------------------------------------
             0.5240   data required time
            -0.7378   data arrival time
---------------------------------------------------------
             0.2138   slack (MET)
```

Corner 3 of 16: ff_n40C_1v56

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ff_n40C_1v56.lib

===== SETUP SUMMARY =====
worst slack max 5.89
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2737    0.2737 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3333    0.6070 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2114    0.8184 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.8200 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.3163    1.1363 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.3165    1.4528 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.2898    1.7426 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.6607    2.4032 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.7219    3.1251 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   0.3388    3.4639 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1381    3.6020 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   0.7138    4.3158 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   0.2614    4.5772 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.1403    4.7176 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   0.8091    5.5267 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.2661    5.7928 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002    5.7930 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             5.7930   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.2737   11.2737 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3681   11.6418 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2220   11.8638 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009   11.8648 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.8648   clock reconvergence pessimism
  -0.1802   11.6845   library setup time
            11.6845   data required time
---------------------------------------------------------
            11.6845   data required time
            -5.7930   data arrival time
---------------------------------------------------------
             5.8916   slack (MET)



===== HOLD SUMMARY =====
```

```
worst slack min 0.32
tns min 0.00
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2737    0.2737 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3333    0.6070 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2083    0.8153 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.8170 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2973    1.1142 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    1.1143 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             1.1143   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2737    0.2737 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3333    0.6070 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2083    0.8153 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    0.8168 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.8168   clock reconvergence pessimism
  -0.0211    0.7957   library hold time
             0.7957   data required time
---------------------------------------------------------
             0.7957   data required time
            -1.1143   data arrival time
---------------------------------------------------------
             0.3186   slack (MET)
```

Corner 4 of 16: ff_n40C_1v65

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ff_n40C_1v65.lib

===== SETUP SUMMARY =====
worst slack max 6.63
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2459    0.2459 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2925    0.5384 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1853    0.7238 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.7254 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2773    1.0027 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2807    1.2834 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.2500    1.5334 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.5841    2.1176 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.6020    2.7196 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   0.2961    3.0157 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1119    3.1275 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   0.6028    3.7304 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   0.2114    3.9418 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.1246    4.0664 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   0.6985    4.7649 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.2111    4.9760 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002    4.9762 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             4.9762   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.2459   11.2459 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3232   11.5691 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1923   11.7614 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009   11.7624 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.7624   clock reconvergence pessimism
  -0.1527   11.6097   library setup time
            11.6097   data required time
---------------------------------------------------------
            11.6097   data required time
            -4.9762   data arrival time
---------------------------------------------------------
             6.6335   slack (MET)



===== HOLD SUMMARY =====
```

```
worst slack min 0.28
tns min 0.00
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2459    0.2459 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2925    0.5384 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1825    0.7210 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.7226 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2604    0.9830 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    0.9830 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             0.9830   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2459    0.2459 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2925    0.5384 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1825    0.7210 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    0.7224 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.7224   clock reconvergence pessimism
  -0.0195    0.7029   library hold time
             0.7029   data required time
---------------------------------------------------------
             0.7029   data required time
            -0.9830   data arrival time
---------------------------------------------------------
             0.2801   slack (MET)
```

Corner 5 of 16: ff_n40C_1v76

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ff_n40C_1v76.lib

===== SETUP SUMMARY =====
worst slack max 7.28
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2209    0.2209 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2567    0.4776 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1624    0.6400 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.6417 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2431    0.8847 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2489    1.1336 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.2137    1.3474 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.5133    1.8607 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.4979    2.3586 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   0.2590    2.6176 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0901    2.7078 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   0.5079    3.2157 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   0.1671    3.3828 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.1110    3.4938 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   0.6014    4.0951 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.1655    4.2606 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002    4.2608 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             4.2608   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.2209   11.2209 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2837   11.5045 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1670   11.6715 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0010   11.6725 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.6725   clock reconvergence pessimism
  -0.1281   11.5443   library setup time
            11.5443   data required time
---------------------------------------------------------
            11.5443   data required time
            -4.2608   data arrival time
---------------------------------------------------------
             7.2836   slack (MET)



===== HOLD SUMMARY =====
```

```
worst slack min 0.25
tns min 0.00
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2209    0.2209 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2567    0.4776 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1601    0.6377 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.6394 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2282    0.8675 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    0.8676 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             0.8676   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2209    0.2209 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2567    0.4776 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1601    0.6377 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    0.6392 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.6392   clock reconvergence pessimism
  -0.0187    0.6205   library hold time
             0.6205   data required time
---------------------------------------------------------
             0.6205   data required time
            -0.8676   data arrival time
---------------------------------------------------------
             0.2471   slack (MET)
```

Corner 6 of 16: ff_n40C_1v95

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ff_n40C_1v95.lib

===== SETUP SUMMARY =====
worst slack max 8.01
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][31]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.1915    0.1915 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2156    0.4071 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1358    0.5429 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.5445 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.2031    0.7476 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2123    0.9599 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.1705    1.1304 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.4340    1.5644 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.3984    1.9627 v _07716_/X (sky130_fd_sc_hd__mux4_2)
   0.1489    2.1116 ^ _07717_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0813    2.1930 v _07719_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0731    2.2660 ^ _08429_/Y (sky130_fd_sc_hd__nand2_1)
   0.1231    2.3891 v _08436_/Y (sky130_fd_sc_hd__a31oi_1)
   0.2282    2.6173 ^ _08462_/Y (sky130_fd_sc_hd__o211ai_1)
   0.1944    2.8117 v _08477_/Y (sky130_fd_sc_hd__a41oi_1)
   0.5930    3.4047 ^ _08478_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0559    3.4606 v _08480_/Y (sky130_fd_sc_hd__a21oi_1)
   0.0002    3.4608 v core.CPU_Xreg_value_a4[10][31]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             3.4608   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.1915   11.1915 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2319   11.4233 ^ clkbuf_3_4__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1399   11.5633 ^ clkbuf_leaf_75_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0011   11.5644 ^ core.CPU_Xreg_value_a4[10][31]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.5644   clock reconvergence pessimism
  -0.0982   11.4662   library setup time
             11.4662   data required time
---------------------------------------------------------
             11.4662   data required time
             -3.4608   data arrival time
---------------------------------------------------------
              8.0055   slack (MET)
===== HOLD SUMMARY =====
worst slack min 0.21
tns min 0.00
```

```
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.1915    0.1915 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2156    0.4071 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1338    0.5409 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.5426 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.1903    0.7329 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    0.7330 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             0.7330   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.1915    0.1915 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2156    0.4071 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1338    0.5409 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    0.5424 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.5424   clock reconvergence pessimism
  -0.0168    0.5256   library hold time
             0.5256   data required time
---------------------------------------------------------
             0.5256   data required time
            -0.7330   data arrival time
---------------------------------------------------------
             0.2074   slack (MET)
```

Corner 7 of 16: ss_100C_1v40

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_100C_1v40.lib

===== SETUP SUMMARY =====
worst slack max -3.57
tns max -543.03
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.5644    0.5644 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7453    1.3097 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5123    1.8220 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    1.8235 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.9816    2.8052 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.7086    3.5138 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.7632    4.2770 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   1.3638    5.6408 ^ place416/X (sky130_fd_sc_hd__buf_4)
   2.4690    8.1098 v _07716_/X (sky130_fd_sc_hd__mux4_2)
   0.6295    8.7393 ^ _07717_/Y (sky130_fd_sc_hd__mux2i_1)
   0.5068    9.2461 v _07719_/Y (sky130_fd_sc_hd__mux2i_1)
   0.2653    9.5113 ^ _08429_/Y (sky130_fd_sc_hd__nand2_1)
   0.7479   10.2593 v _08436_/Y (sky130_fd_sc_hd__a31oi_1)
   0.9828   11.2421 ^ _08462_/Y (sky130_fd_sc_hd__o211ai_1)
   1.2548   12.4968 v _08477_/Y (sky130_fd_sc_hd__a41oi_1)
   2.3814   14.8783 ^ _08478_/Y (sky130_fd_sc_hd__mux2i_1)
   0.9825   15.8608 v _09004_/Y (sky130_fd_sc_hd__a21oi_1)
   0.0002   15.8610 v core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            15.8610   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.5644   11.5644 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7927   12.3570 ^ clkbuf_3_4__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5471   12.9041 ^ clkbuf_leaf_77_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0011   12.9052 ^ core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   12.9052   clock reconvergence pessimism
  -0.6120   12.2933   library setup time
            12.2933   data required time
---------------------------------------------------------
            12.2933   data required time
           -15.8610   data arrival time
---------------------------------------------------------
            -3.5677   slack (VIOLATED)
===== HOLD SUMMARY =====
worst slack min 0.95
tns min 0.00
```

```
Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.5644    0.5644 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7771    1.3415 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5357    1.8772 ^ clkbuf_leaf_55_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    1.8781 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.8489    2.7270 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    2.7271 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
             2.7271   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.5644    0.5644 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7771    1.3415 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5357    1.8772 ^ clkbuf_leaf_55_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0010    1.8782 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
   0.0000    1.8782   clock reconvergence pessimism
  -0.1020    1.7762   library hold time
             1.7762   data required time
---------------------------------------------------------
             1.7762   data required time
            -2.7271   data arrival time
---------------------------------------------------------
             0.9509   slack (MET)
```

Corner 8 of 16: ss_100C_1v60

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_100C_1v60.lib

===== SETUP SUMMARY =====
worst slack max 1.14
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.4201    0.4201 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5372    0.9573 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3649    1.3223 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    1.3239 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.6670    1.9908 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.5152    2.5060 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.5300    3.0360 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.9909    4.0270 ^ place416/X (sky130_fd_sc_hd__buf_4)
   1.5827    5.6097 v _07716_/X (sky130_fd_sc_hd__mux4_2)
   0.4318    6.0415 ^ _07717_/Y (sky130_fd_sc_hd__mux2i_1)
   0.3414    6.3829 v _07719_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1868    6.5696 ^ _08429_/Y (sky130_fd_sc_hd__nand2_1)
   0.5258    7.0954 v _08436_/Y (sky130_fd_sc_hd__a31oi_1)
   0.6604    7.7559 ^ _08462_/Y (sky130_fd_sc_hd__o211ai_1)
   0.8617    8.6175 v _08477_/Y (sky130_fd_sc_hd__a41oi_1)
   1.6423   10.2598 ^ _08478_/Y (sky130_fd_sc_hd__mux2i_1)
   0.5998   10.8596 v _09004_/Y (sky130_fd_sc_hd__a21oi_1)
   0.0002   10.8598 v core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            10.8598   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.4201   11.4201 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5704   11.9905 ^ clkbuf_3_4__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3878   12.3783 ^ clkbuf_leaf_77_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0012   12.3794 ^ core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   12.3794   clock reconvergence pessimism
  -0.3769   12.0025   library setup time
            12.0025   data required time
---------------------------------------------------------
            12.0025   data required time
           -10.8598   data arrival time
---------------------------------------------------------
             1.1427   slack (MET)
===== HOLD SUMMARY =====
worst slack min 0.67
tns min 0.00
```

```
Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.4201    0.4201 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5601    0.9802 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3803    1.3605 ^ clkbuf_leaf_55_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    1.3614 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.5979    1.9593 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    1.9594 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
             1.9594   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.4201    0.4201 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5601    0.9802 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3803    1.3605 ^ clkbuf_leaf_55_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0010    1.3615 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
   0.0000    1.3615   clock reconvergence pessimism
  -0.0767    1.2848   library hold time
             1.2848   data required time
---------------------------------------------------------
             1.2848   data required time
            -1.9594   data arrival time
---------------------------------------------------------
             0.6746   slack (MET)
```

Corner 9 of 16: ss_n40C_1v28

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_n40C_1v28.lib

===== SETUP SUMMARY =====
worst slack max -31.43
tns max -15435.67
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   1.3572    1.3572 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   2.0156    3.3728 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.2882    4.6610 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    4.6625 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.4382    7.1007 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.7588    8.8594 ^ place469/X (sky130_fd_sc_hd__buf_4)
   2.1222   10.9816 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   3.7441   14.7257 ^ place416/X (sky130_fd_sc_hd__buf_4)
   7.2840   22.0097 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   2.3212   24.3309 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   1.7423   26.0732 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   5.6870   31.7602 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   3.9388   35.6990 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   1.0353   36.7343 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   5.4465   42.1808 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   3.4192   45.6000 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002   45.6002 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            45.6002   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   1.3572   12.3572 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   2.2154   14.5726 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.4140   15.9866 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0008   15.9874 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   15.9874   clock reconvergence pessimism
  -1.8198   14.1676   library setup time
            14.1676   data required time
---------------------------------------------------------
            14.1676   data required time
           -45.6002   data arrival time
---------------------------------------------------------
           -31.4325   slack (VIOLATED)


===== HOLD SUMMARY =====
```

```
worst slack min 1.85
tns min 0.00
Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   1.3572    1.3572 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   2.1070    3.4642 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.3574    4.8216 ^ clkbuf_leaf_41_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    4.8230 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.7285    6.5515 v core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    6.5516 v core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             6.5516   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   1.3572    1.3572 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   2.1611    3.5183 ^ clkbuf_3_3__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.4118    4.9301 ^ clkbuf_leaf_43_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    4.9310 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    4.9310   clock reconvergence pessimism
  -0.2281    4.7028   library hold time
             4.7028   data required time
---------------------------------------------------------
             4.7028   data required time
            -6.5516   data arrival time
---------------------------------------------------------
             1.8488   slack (MET)
```

Corner 10 of 16: ss_n40C_1v35

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_n40C_1v35.lib

===== SETUP SUMMARY =====
worst slack max -17.65
tns max -7396.50
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.9793    0.9793 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.4242    2.4035 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.9144    3.3179 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    3.3194 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.6852    5.0046 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.2746    6.2791 ^ place469/X (sky130_fd_sc_hd__buf_4)
   1.4677    7.7468 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   2.7036   10.4504 ^ place416/X (sky130_fd_sc_hd__buf_4)
   4.7555   15.2059 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   1.6578   16.8637 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   1.1376   18.0013 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   3.7805   21.7819 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   2.4833   24.2651 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.7467   25.0118 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   3.7917   28.8035 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   2.1639   30.9674 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002   30.9676 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            30.9676   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.9793   11.9793 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.5627   13.5420 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.9986   14.5406 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0008   14.5415 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   14.5415   clock reconvergence pessimism
  -1.2227   13.3188   library setup time
            13.3188   data required time
---------------------------------------------------------
            13.3188   data required time
           -30.9676   data arrival time
---------------------------------------------------------
           -17.6488   slack (VIOLATED)



===== HOLD SUMMARY =====
```

```
worst slack min 1.38
tns min 0.00
Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.9793    0.9793 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.4889    2.4682 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.9629    3.4312 ^ clkbuf_leaf_41_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    3.4326 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.2722    4.7048 v core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    4.7049 v core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             4.7049   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.9793    0.9793 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.5264    2.5057 ^ clkbuf_3_3__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.0001    3.5058 ^ clkbuf_leaf_43_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    3.5067 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    3.5067   clock reconvergence pessimism
  -0.1815    3.3252   library hold time
             3.3252   data required time
---------------------------------------------------------
             3.3252   data required time
            -4.7049   data arrival time
---------------------------------------------------------
             1.3796   slack (MET)
```

Corner 11 of 16: ss_n40C_1v40

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_n40C_1v40.lib

===== SETUP SUMMARY =====
worst slack max -11.95
tns max -3993.60
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.8153    0.8153 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.1584    1.9738 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7473    2.7211 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    2.7226 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.3609    4.0835 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.0581    5.1416 ^ place469/X (sky130_fd_sc_hd__buf_4)
   1.1981    6.3397 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   2.2289    8.5686 ^ place416/X (sky130_fd_sc_hd__buf_4)
   3.7584   12.3269 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   1.3455   13.6724 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.8871   14.5595 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   2.9904   17.5500 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   1.9618   19.5118 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.6271   20.1389 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   3.0556   23.1945 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   1.6934   24.8879 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002   24.8881 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            24.8881   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.8153   11.8153 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.2713   13.0866 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.8120   13.8986 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0008   13.8995 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   13.8995   clock reconvergence pessimism
  -0.9572   12.9423   library setup time
            12.9423   data required time
---------------------------------------------------------
            12.9423   data required time
           -24.8881   data arrival time
---------------------------------------------------------
           -11.9458   slack (VIOLATED)



===== HOLD SUMMARY =====
```

```
worst slack min 1.16
tns min 0.00
Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.8153    0.8153 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.2117    2.0270 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7849    2.8119 ^ clkbuf_leaf_41_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    2.8133 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.0629    3.8763 v core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    3.8764 v core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             3.8764   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.8153    0.8153 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.2421    2.0575 ^ clkbuf_3_3__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.8137    2.8712 ^ clkbuf_leaf_43_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    2.8721 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    2.8721   clock reconvergence pessimism
  -0.1523    2.7198   library hold time
             2.7198   data required time
---------------------------------------------------------
             2.7198   data required time
            -3.8764   data arrival time
---------------------------------------------------------
             1.1566   slack (MET)
```

Corner 12 of 16: ss_n40C_1v44

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_n40C_1v44.lib

===== SETUP SUMMARY =====
worst slack max -8.49
tns max -2218.38
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.7157    0.7157 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.0060    1.7217 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.6516    2.3733 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    2.3748 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.1694    3.5442 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.9250    4.4692 ^ place469/X (sky130_fd_sc_hd__buf_4)
   1.0218    5.4910 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   1.9230    7.4140 ^ place416/X (sky130_fd_sc_hd__buf_4)
   3.1751   10.5892 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   1.1553   11.7445 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.7525   12.4970 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   2.5303   15.0272 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   1.6188   16.6461 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.5508   17.1969 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   2.6328   19.8296 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   1.3915   21.2211 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002   21.2213 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            21.2213   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.7157   11.7157 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.1037   12.8194 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.6991   13.5185 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0008   13.5193 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   13.5193   clock reconvergence pessimism
  -0.7911   12.7282   library setup time
            12.7282   data required time
---------------------------------------------------------
            12.7282   data required time
           -21.2213   data arrival time
---------------------------------------------------------
            -8.4931   slack (VIOLATED)



===== HOLD SUMMARY =====
```

```
worst slack min 1.02
tns min 0.00
Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.7157    0.7157 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.0523    1.7679 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.6824    2.4503 ^ clkbuf_leaf_41_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    2.4517 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.9336    3.3853 v core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    3.3854 v core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             3.3854   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.7157    0.7157 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   1.0784    1.7941 ^ clkbuf_3_3__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7036    2.4976 ^ clkbuf_leaf_43_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    2.4985 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    2.4985   clock reconvergence pessimism
  -0.1367    2.3619   library hold time
             2.3619   data required time
---------------------------------------------------------
             2.3619   data required time
            -3.3854   data arrival time
---------------------------------------------------------
             1.0235   slack (MET)
```

Corner 13 of 16: ss_n40C_1v60

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_n40C_1v60.lib

===== SETUP SUMMARY =====
worst slack max -0.98
tns max -16.12
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.4901    0.4901 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.6581    1.1482 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.4273    1.5756 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    1.5771 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.7369    2.3140 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.6207    2.9347 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.6562    3.5909 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   1.2700    4.8609 ^ place416/X (sky130_fd_sc_hd__buf_4)
   1.8832    6.7441 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   0.7404    7.4844 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.4319    7.9163 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   1.5203    9.4366 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   0.9409   10.3776 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.3714   10.7490 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   1.6554   12.4044 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.7903   13.1947 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002   13.1949 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
            13.1949   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.4901   11.4901 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7217   12.2119 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.4572   12.6690 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009   12.6699 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   12.6699   clock reconvergence pessimism
  -0.4558   12.2141   library setup time
            12.2141   data required time
---------------------------------------------------------
            12.2141   data required time
           -13.1949   data arrival time
---------------------------------------------------------
            -0.9808   slack (VIOLATED)



===== HOLD SUMMARY =====
```

```
worst slack min 0.69
tns min 0.00
Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.4901    0.4901 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.6889    1.1791 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.4459    1.6249 ^ clkbuf_leaf_41_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    1.6264 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.6276    2.2540 v core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    2.2542 v core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             2.2542   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.4901    0.4901 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.7059    1.1961 ^ clkbuf_3_3__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.4600    1.6560 ^ clkbuf_leaf_43_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    1.6570 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    1.6570   clock reconvergence pessimism
  -0.0944    1.5626   library hold time
             1.5626   data required time
---------------------------------------------------------
             1.5626   data required time
            -2.2542   data arrival time
---------------------------------------------------------
             0.6916   slack (MET)
```

Corner 14 of 16: ss_n40C_1v76

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__ss_n40C_1v76.lib

===== SETUP SUMMARY =====
worst slack max 2.58
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.3785    0.3785 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.4897    0.8682 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3212    1.1894 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    1.1910 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.5320    1.7230 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.4670    2.1900 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.4704    2.6604 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.9394    3.5998 ^ place416/X (sky130_fd_sc_hd__buf_4)
   1.2683    4.8681 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   0.5399    5.4080 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.2966    5.7046 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   1.0630    6.7676 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   0.6241    7.3917 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.2815    7.6732 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   1.1970    8.8702 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.5048    9.3750 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002    9.3752 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             9.3752   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.3785   11.3785 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5367   11.9152 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3375   12.2526 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009   12.2535 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   12.2535   clock reconvergence pessimism
  -0.2966   11.9570   library setup time
            11.9570   data required time
---------------------------------------------------------
            11.9570   data required time
            -9.3752   data arrival time
---------------------------------------------------------
             2.5818   slack (MET)



===== HOLD SUMMARY =====
```

```
worst slack min 0.52
tns min 0.00
Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.3785    0.3785 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5130    0.8914 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3352    1.2267 ^ clkbuf_leaf_55_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009    1.2276 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.4684    1.6960 v core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    1.6961 v core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_4)
             1.6961   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.3785    0.3785 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.5130    0.8914 ^ clkbuf_3_6__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3352    1.2267 ^ clkbuf_leaf_55_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0010    1.2277 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_4)
   0.0000    1.2277   clock reconvergence pessimism
  -0.0493    1.1784   library hold time
             1.1784   data required time
---------------------------------------------------------
             1.1784   data required time
            -1.6961   data arrival time
---------------------------------------------------------
             0.5177   slack (MET)
```

Corner 15 of 16: tt_025C_1v80

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__tt_025C_1v80.lib

===== SETUP SUMMARY =====
worst slack max 6.15
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2597    0.2597 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3138    0.5735 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2050    0.7785 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.7801 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.3248    1.1049 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2988    1.4037 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.2797    1.6834 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.5924    2.2758 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.6699    2.9457 v _07564_/X (sky130_fd_sc_hd__mux4_2)
   0.3282    3.2739 ^ _07568_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1610    3.4349 v _08312_/Y (sky130_fd_sc_hd__o22ai_1)
   0.5997    4.0346 ^ _08315_/Y (sky130_fd_sc_hd__o311ai_1)
   0.3259    4.3605 ^ _08316_/X (sky130_fd_sc_hd__o31a_1)
   0.1904    4.5509 v _08317_/Y (sky130_fd_sc_hd__o22ai_1)
   0.7146    5.2655 ^ _08319_/Y (sky130_fd_sc_hd__o211ai_1)
   0.2468    5.5123 v _09183_/Y (sky130_fd_sc_hd__o21ai_0)
   0.0002    5.5125 v core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             5.5125   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.2597   11.2597 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3436   11.6033 ^ clkbuf_3_5__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2131   11.8164 ^ clkbuf_leaf_81_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0009   11.8174 ^ core.CPU_Xreg_value_a4[27][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.8174   clock reconvergence pessimism
  -0.1597   11.6577   library setup time
            11.6577   data required time
---------------------------------------------------------
            11.6577   data required time
            -5.5125   data arrival time
---------------------------------------------------------
             6.1451   slack (MET)



===== HOLD SUMMARY =====
```

```
worst slack min 0.33
tns min 0.00
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2597    0.2597 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3138    0.5735 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2021    0.7756 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0016    0.7772 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.3068    1.0841 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    1.0841 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             1.0841   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2597    0.2597 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3138    0.5735 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2021    0.7756 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0014    0.7771 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.7771   clock reconvergence pessimism
  -0.0278    0.7493   library hold time
             0.7493   data required time
---------------------------------------------------------
             0.7493   data required time
            -1.0841   data arrival time
---------------------------------------------------------
             0.3349   slack (MET)
```

Corner 16 of 16: tt_100C_1v80

```
OpenROAD v2.0-25794-g7cbd77e248
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
CORNER_LIBRARY: /home/vscode/skywater-pdk-libs-sky130_fd_sc_hd/timing/sky130_fd_sc_hd__tt_100C_1v80.lib

===== SETUP SUMMARY =====
worst slack max 6.24
tns max 0.00
Startpoint: core.CPU_is_slli_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2550    0.2550 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3045    0.5595 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2013    0.7608 ^ clkbuf_leaf_60_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.7624 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.3254    1.0878 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.2906    1.3784 ^ place469/X (sky130_fd_sc_hd__buf_4)
   0.2714    1.6498 ^ _07381_/Y (sky130_fd_sc_hd__mux2i_4)
   0.5637    2.2135 ^ place416/X (sky130_fd_sc_hd__buf_4)
   0.6795    2.8929 v _07716_/X (sky130_fd_sc_hd__mux4_2)
   0.2249    3.1178 ^ _07717_/Y (sky130_fd_sc_hd__mux2i_1)
   0.1649    3.2827 v _07719_/Y (sky130_fd_sc_hd__mux2i_1)
   0.0955    3.3783 ^ _08429_/Y (sky130_fd_sc_hd__nand2_1)
   0.2540    3.6323 v _08436_/Y (sky130_fd_sc_hd__a31oi_1)
   0.3216    3.9539 ^ _08462_/Y (sky130_fd_sc_hd__o211ai_1)
   0.3996    4.3535 v _08477_/Y (sky130_fd_sc_hd__a41oi_1)
   0.8457    5.1992 ^ _08478_/Y (sky130_fd_sc_hd__mux2i_1)
   0.2022    5.4014 v _09004_/Y (sky130_fd_sc_hd__a21oi_1)
   0.0002    5.4017 v core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
             5.4017   data arrival time

  11.0000   11.0000   clock clk (rise edge)
   0.0000   11.0000   clock source latency
   0.0000   11.0000 ^ pll/CLK (avsdpll)
   0.2550   11.2550 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3245   11.5795 ^ clkbuf_3_4__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.2133   11.7928 ^ clkbuf_leaf_77_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0012   11.7940 ^ core.CPU_Xreg_value_a4[1][31]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000   11.7940   clock reconvergence pessimism
  -0.1558   11.6382   library setup time
             11.6382   data required time
---------------------------------------------------------
             11.6382   data required time
             -5.4017   data arrival time
---------------------------------------------------------
              6.2365   slack (MET)
===== HOLD SUMMARY =====
worst slack min 0.34
tns min 0.00
```

```
Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Delay      Time   Description
---------------------------------------------------------
   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2550    0.2550 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3045    0.5595 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1986    0.7581 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0017    0.7598 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.3082    1.0679 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.0001    1.0680 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
             1.0680   data arrival time

   0.0000    0.0000   clock clk (rise edge)
   0.0000    0.0000   clock source latency
   0.0000    0.0000 ^ pll/CLK (avsdpll)
   0.2550    0.2550 ^ clkbuf_0_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.3045    0.5595 ^ clkbuf_3_7__f_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.1986    0.7581 ^ clkbuf_leaf_57_CLK/X (sky130_fd_sc_hd__clkbuf_16)
   0.0015    0.7596 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.0000    0.7596   clock reconvergence pessimism
  -0.0301    0.7294   library hold time
             0.7294   data required time
---------------------------------------------------------
             0.7294   data required time
            -1.0680   data arrival time
---------------------------------------------------------
             0.3385   slack (MET)
```